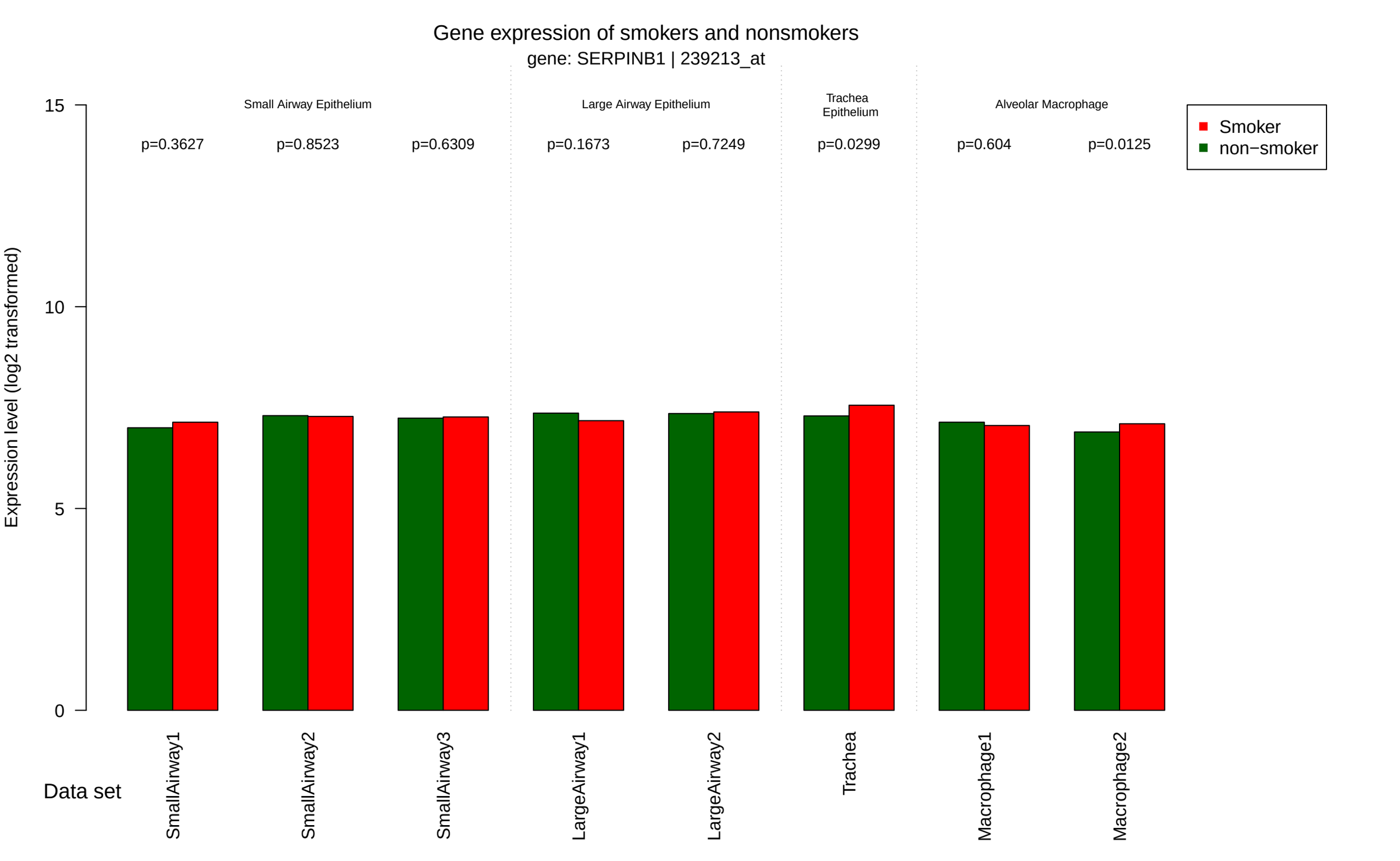
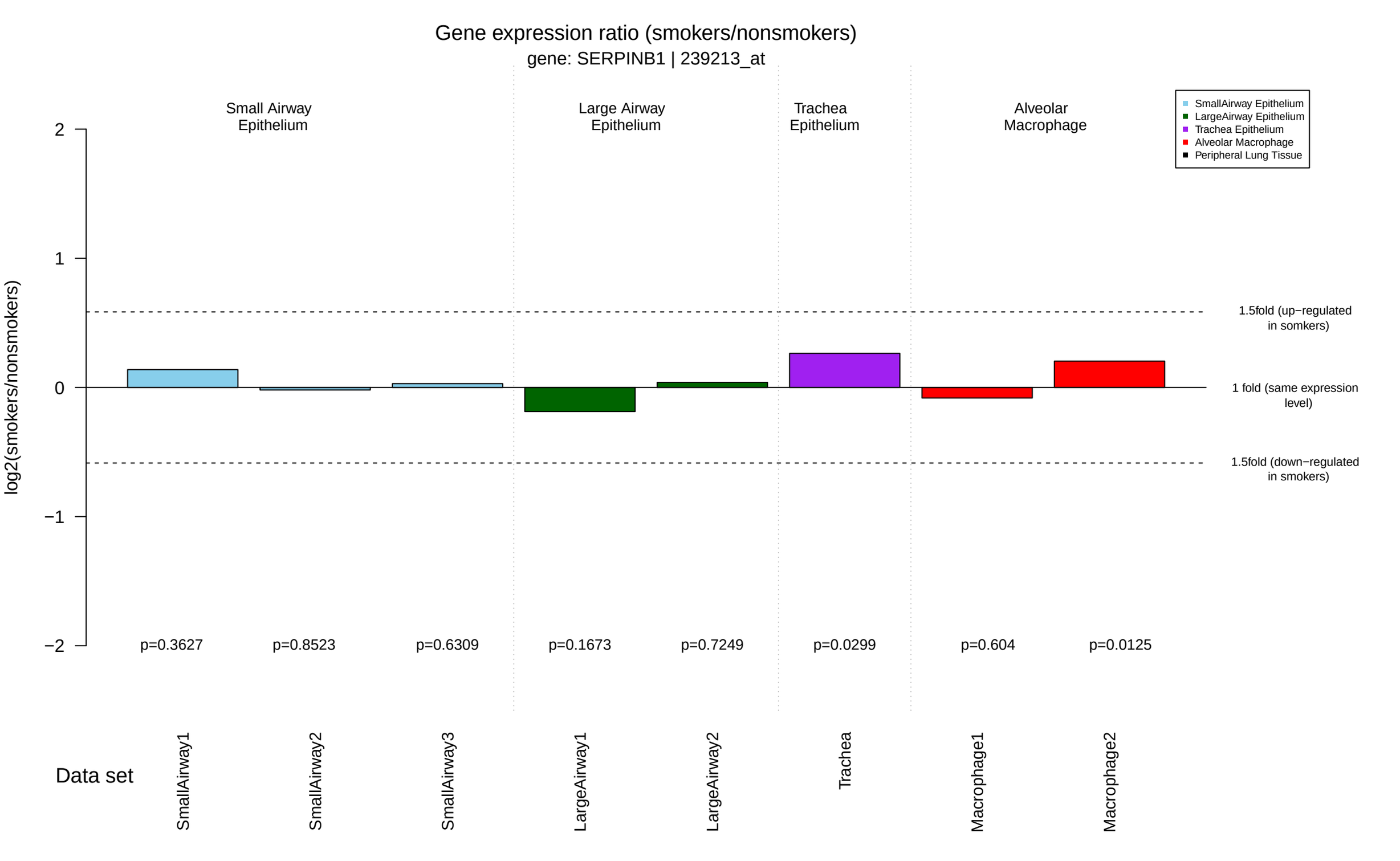
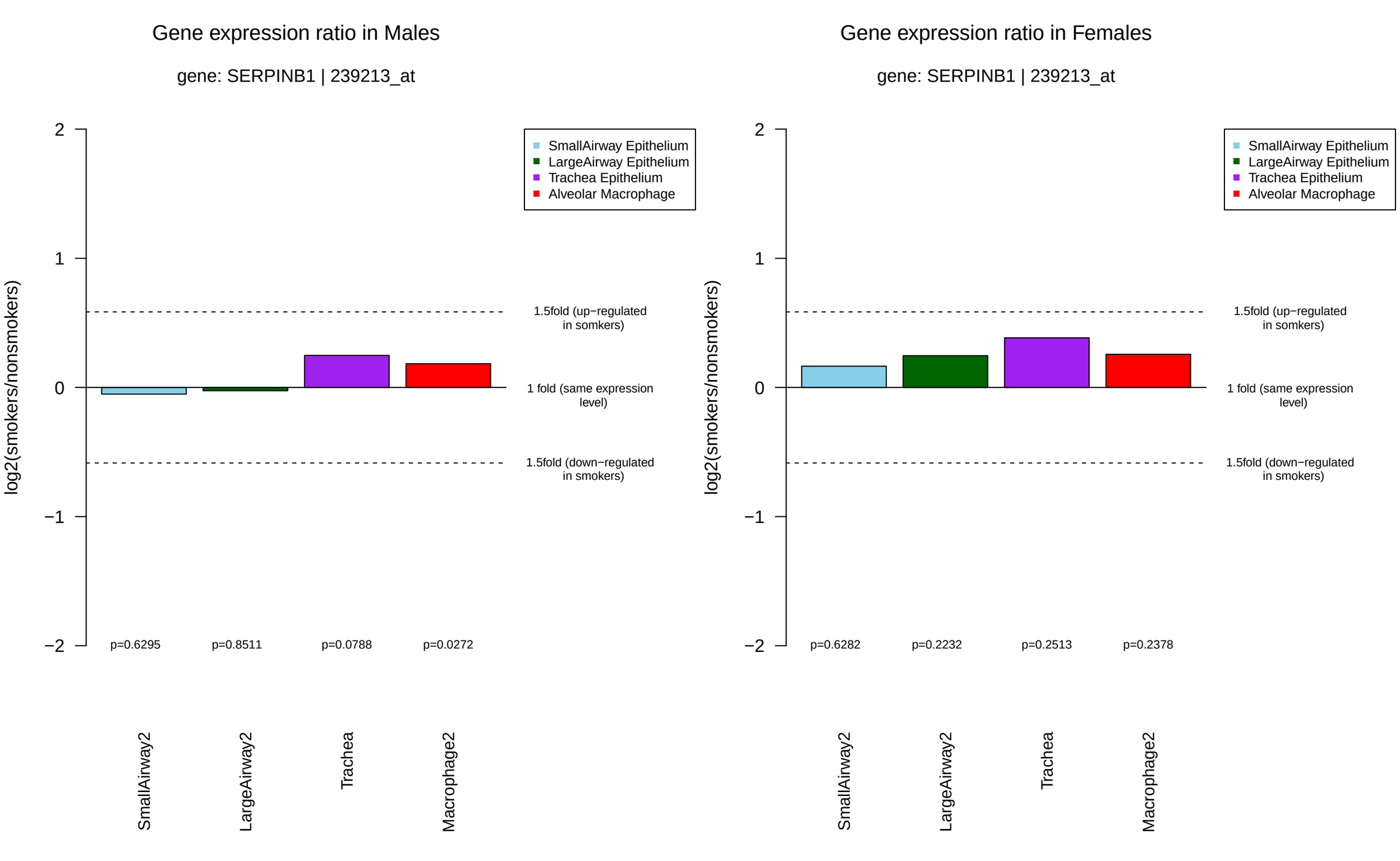
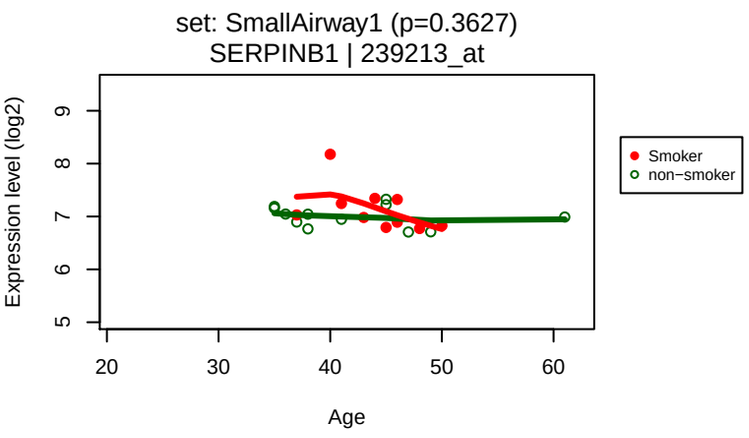
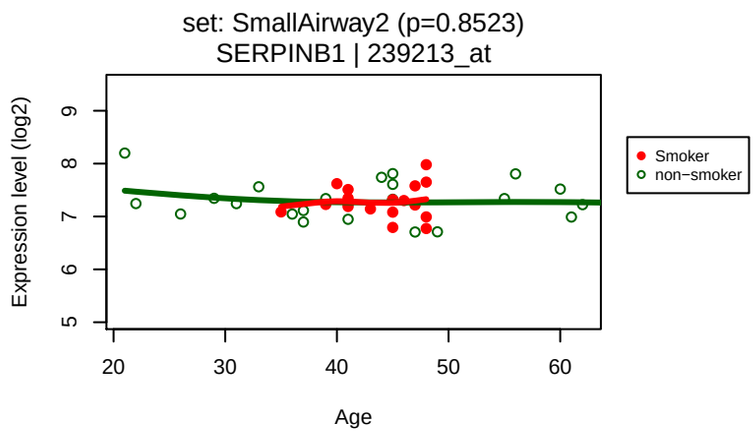
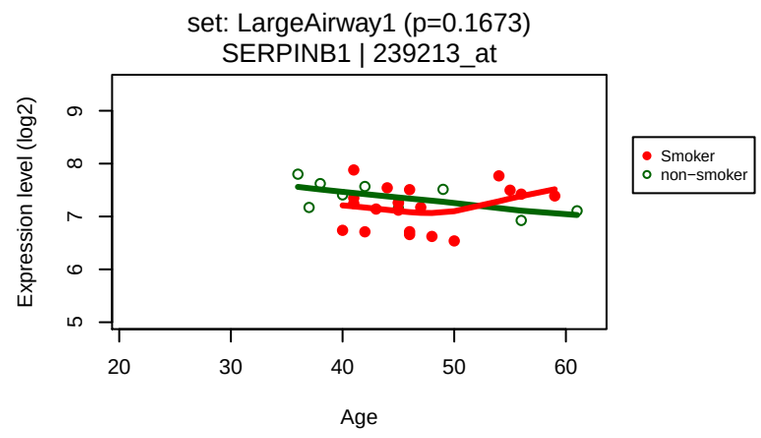

Gene expression of smokers and nonsmokers
gene: SERPINB1 | 239213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3627
p=0.8523
p=0.6309
p=0.1673
p=0.7249
p=0.0299
p=0.604
p=0.0125
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

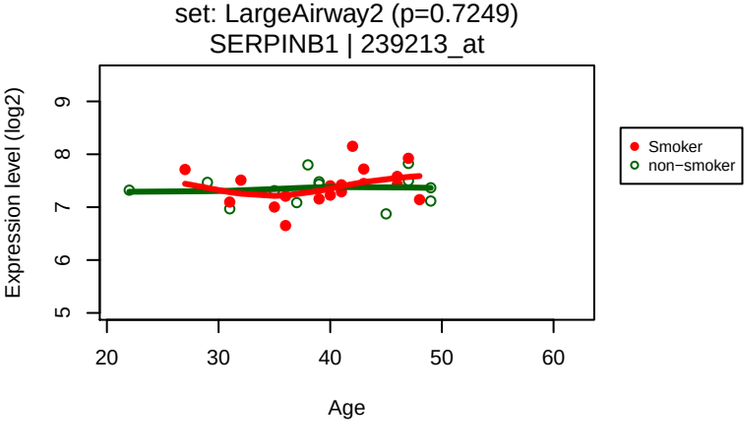

Gene expression ratio (smokers/nonsmokers)
gene: SERPINB1 | 239213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3627
p=0.8523
p=0.6309
p=0.1673
p=0.7249
p=0.0299
p=0.604
p=0.0125
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

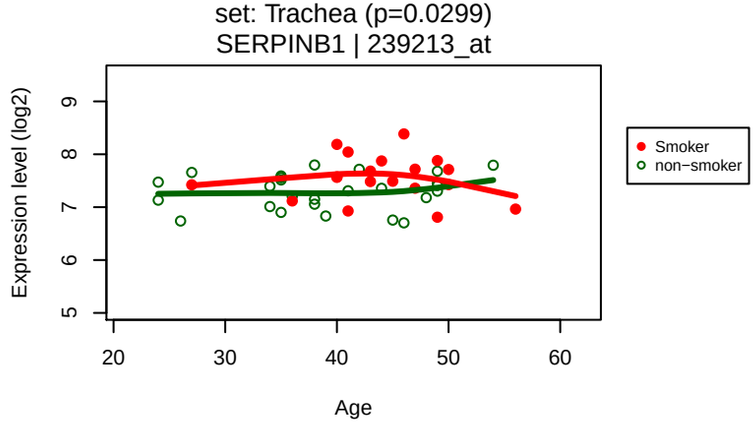

Gene expression ratio in Males
gene: SERPINB1 | 239213_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6295
p=0.8511
p=0.0788
p=0.0272
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINB1 | 239213_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6282
p=0.2232
p=0.2513
p=0.2378
SmallAirway2
LargeAirway2
Trachea
Macrophage2

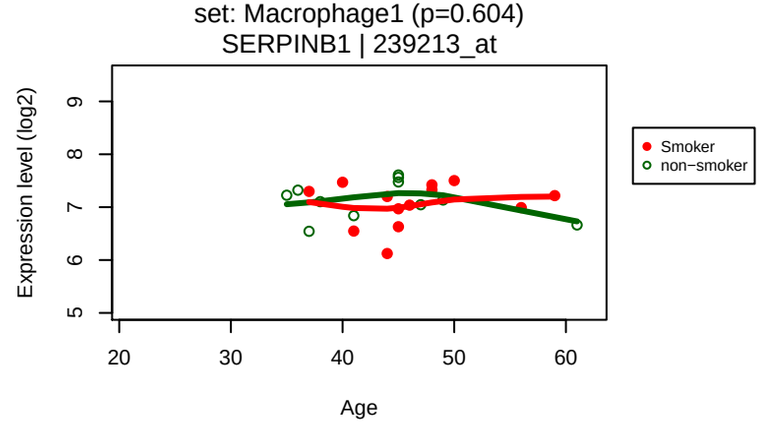

set: SmallAirway1 (p=0.3627)
SERPINB1 | 239213_at
Expression level (log2)
Age
Smoker
non−smoker

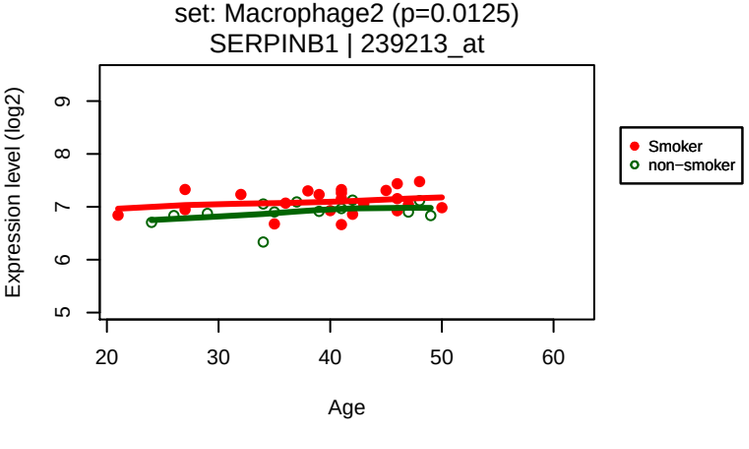

set: SmallAirway2 (p=0.8523)
SERPINB1 | 239213_at
Expression level (log2)
Age
Smoker
non−smoker

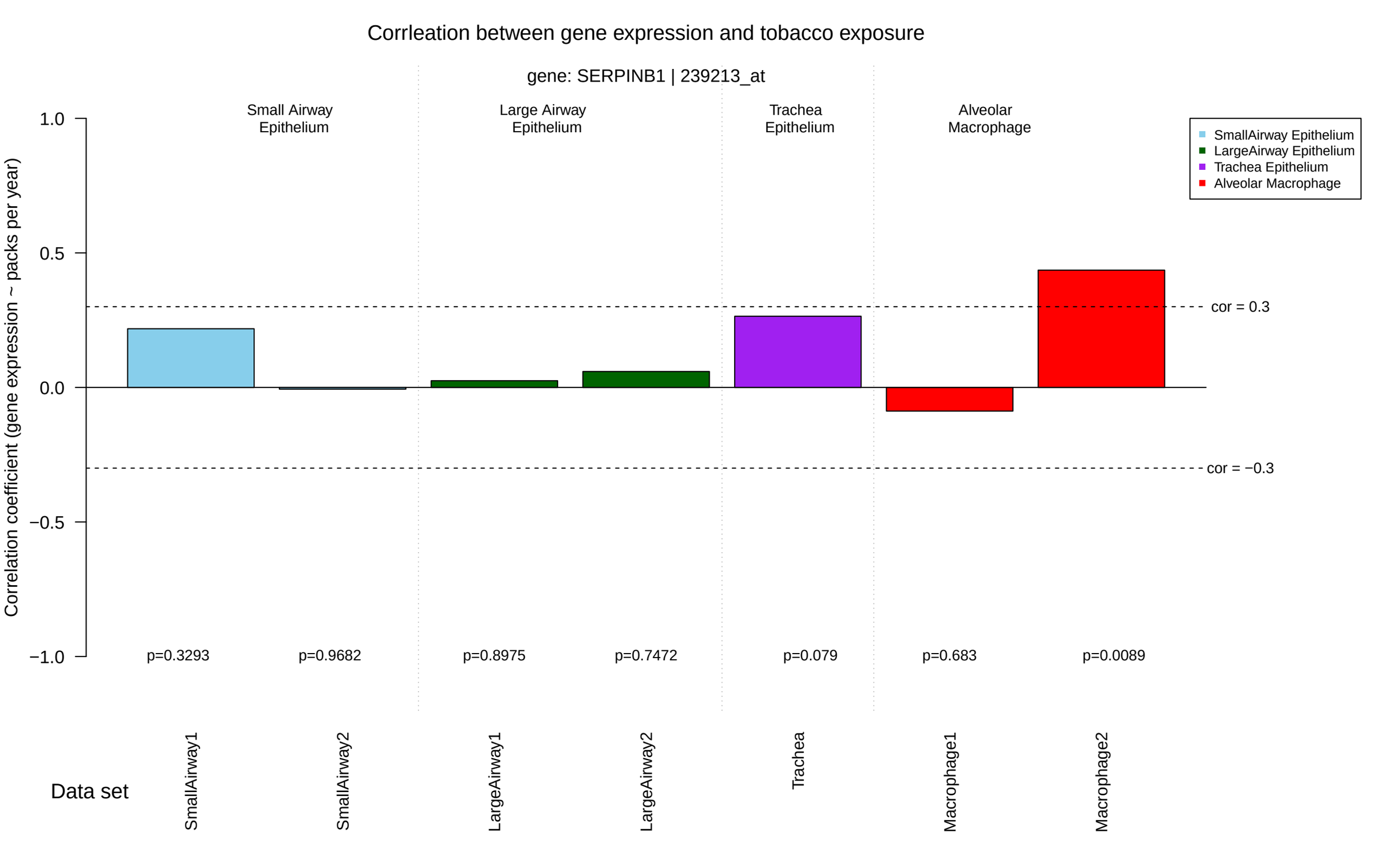

set: LargeAirway1 (p=0.1673)
SERPINB1 | 239213_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7249)
SERPINB1 | 239213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0299)
SERPINB1 | 239213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.604)
SERPINB1 | 239213_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0125)
SERPINB1 | 239213_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINB1 | 239213_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3293
p=0.9682
p=0.8975
p=0.7472
p=0.079
p=0.683
p=0.0089
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2